

Lab: Service Mesh

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Lab: 4

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Task -1 – Prepare and Verify your environment

```
> kubectl get nodes
```

NAME	STATUS	ROLES	AGE	VERSION
ip-172-31-2-250.ec2.internal	Ready	<none>	2m39s	v1.35.5-eks-3385e9b
ip-172-31-32-161.ec2.internal	Ready	<none>	2m38s	v1.35.5-eks-3385e9b

```
> kubectl --namespace kube-system get pods
```

NAME	READY	STATUS	RESTARTS	AGE
aws-node-lvc5q	2/2	Running	0	88s
aws-node-zx9m8	2/2	Running	0	88s
coredns-7fc5967d79-6m9q6	1/1	Running	0	58s
coredns-7fc5967d79-t6x5z	1/1	Running	0	58s
eks-node-monitoring-agent-597kg	1/1	Running	2 (53s ago)	3m8s
eks-node-monitoring-agent-mbkds	1/1	Running	2 (53s ago)	3m9s
eks-pod-identity-agent-5dvtb	1/1	Running	0	59s
eks-pod-identity-agent-zr49f	1/1	Running	0	58s
kube-proxy-jknvt	1/1	Running	0	56s
kube-proxy-l4kkq	1/1	Running	0	56s
metrics-server-b8c45bcb8-4kpzg	1/1	Running	1 (44s ago)	4m42s
metrics-server-b8c45bcb8-gbq5d	1/1	Running	1 (44s ago)	4m42s

Task 0 – Install Istio Control Plane

```
> kubectl -n istio-system get pods
```

NAME	READY	STATUS	RESTARTS	AGE
istio-egressgateway-64d76c9487-b9brh	1/1	Running	0	63s
istio-ingressgateway-7c4fcf8489-wfdfh	1/1	Running	0	63s
istiod-6ffcf45f8b-xjnsb	1/1	Running	0	75s

```
> kubectl -n istio-system get svc
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	AGE
istio-egressgateway	ClusterIP	10.100.141.251	<none>	84s
istio-ingressgateway	LoadBalancer	10.100.8.69		

```
a32f99e6bb7fe40e8a83241aca41ac9c-511683434.us-east-1.elb.amazonaws.com
15021:31302/TCP,80:31322/TCP,443:30301/TCP,31400:32579/TCP,15443:31973/TCP 84s
istiod ClusterIP 10.100.128.79 <none>
15010/TCP,15012/TCP,443/TCP,15014/TCP 96s
istiod-revision-tag-default ClusterIP 10.100.183.253 <none>
15010/TCP,15012/TCP,443/TCP,15014/TCP 68s
```

```
> kubectl -n istio-system get svc istio-ingressgateway
NAME TYPE CLUSTER-IP EXTERNAL-IP AGE
PORT(S)
istio-ingressgateway LoadBalancer 10.100.8.69
a32f99e6bb7fe40e8a83241aca41ac9c-511683434.us-east-1.elb.amazonaws.com
15021:31302/TCP,80:31322/TCP,443:30301/TCP,31400:32579/TCP,15443:31973/TCP 119s
```

Task 1 – Deploy test application

```
> kubectl -n default get pods
NAME READY STATUS RESTARTS AGE
adservice-v1-565cfbcd84-dgjmkn 2/2 Running 0 63s
cartservice-v1-565f7849cf-7vr14 2/2 Running 2 (49s ago) 68s
checkoutservice-v1-84dd58c487-7sq56 2/2 Running 0 74s
currencyservice-v1-9c7bc475-7td2w 2/2 Running 0 67s
emailservice-v1-cbb949b65-psch5 2/2 Running 0 75s
emailservice-v2-7cccbc555b-577k6 2/2 Running 0 75s
frontend-v1-777cb949cc-ktv6k 2/2 Running 0 72s
loadgenerator-549975cd77-s7t1l 2/2 Running 3 (27s ago) 67s
paymentservice-v1-6bfb64b4b8-t2r47 2/2 Running 0 71s
productcatalogservice-v1-b9c85f4b9-zmtjc 2/2 Running 0 70s
productcatalogservice-v2-6d6bf7bd86-qwgjl 2/2 Running 0 69s
recommendationservice-v1-755664dd9f-f9sw2 2/2 Running 0 73s
redis-cart-v1-664db6988f-7r5j9 2/2 Running 0 63s
shippingservice-v1-868c485955-b2mhj 2/2 Running 0 66s
shippingservice-v2-696b876748-7rw8q 2/2 Running 0 65s
shippingservice-v3-569458c949-ztx7z 2/2 Running 0 65s
```

Task 2 – Observability, service metrics

- How many ops are being currently processed by the application?
33.1 ops/s
- What is the global success rate?
100%
- Are there any request errors occurring in the application at the moment? How do you check that?
0 ops/s (4xxs + 5xxs)
- Which application service processes the most requests? What is its p99 latency and success rate?
currencyservice.default.svc.cluster.local (4.84 ms, 100.00%)

5. What is the current CPU and memory usage by Istio Pilot?

131 MiB Resident Memory

0.00227 CPU Discovery (container)

6. Does Istio Pilot encounters any errors at the moment?

0 Pilot Errors

Task 3 – Observability, distributed tracing

There is only **jaeger-query** in Service dropdown.

1. When the request started (date, time)?

Today, 6:38:11 pm

2. Total request duration?

2.07ms

3. Request protocol, method and status code?

Protocol: HTTP (**net/http** component)

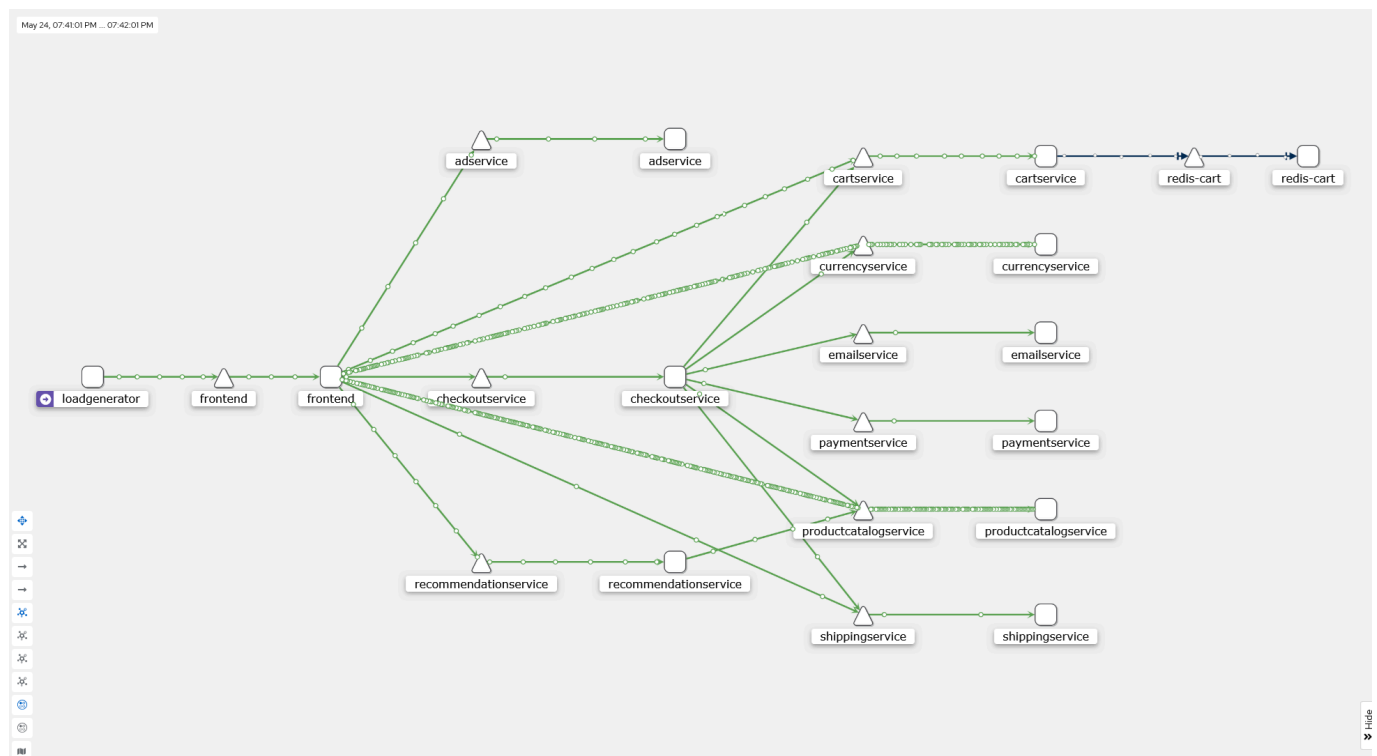
Method: GET

Status Code: 200

4. Duration of nested span?

No nested spans

Task 4 – Observability, mesh topology



1. Is the inter-service communication healthy? How do you check that?

Yes, all connections are green

2. What is the entry-point service of the application?

frontend service

3. Which services communicate with the currency service?

frontend, **checkoutservice**

4. Which services communicate with the productcatalog service?
`frontend, checkoutservice, recommendationservice`
5. Which services does the checkout service communicate with?
`cartservice, currencyservice, emailservice, paymentservice, productcatalogservice, shippingservice`
6. What is the request volume (ops) for currency service? What are its success and error rates?
RPS: 7.07 (100.00% Success, 0.00% Errors)
7. What is the request volume (ops) for payment service? What are its success and error rates?
RPS: 0.13 (100.00% Success, 0.00% Errors)
8. What protocol is used on the communication path between the frontend and the recommendation service?
`grpc`
9. What protocol is used on the communication path between the cart service and the redis service?
`tcp`

Task 5 – Traffic management, ingress gateway

- VirtualService Policy
"First, the policy selects the gateway to configure (frontend-gateway). Then, it configures the gateway to route the entire HTTP traffic to the frontend service onto port 80."
- Gateway Policy
"Opens port 80 for HTTP traffic for all destination services (Hosts: *) discovered in the mesh. The Selector binds the configuration to the specific Ingress instance."

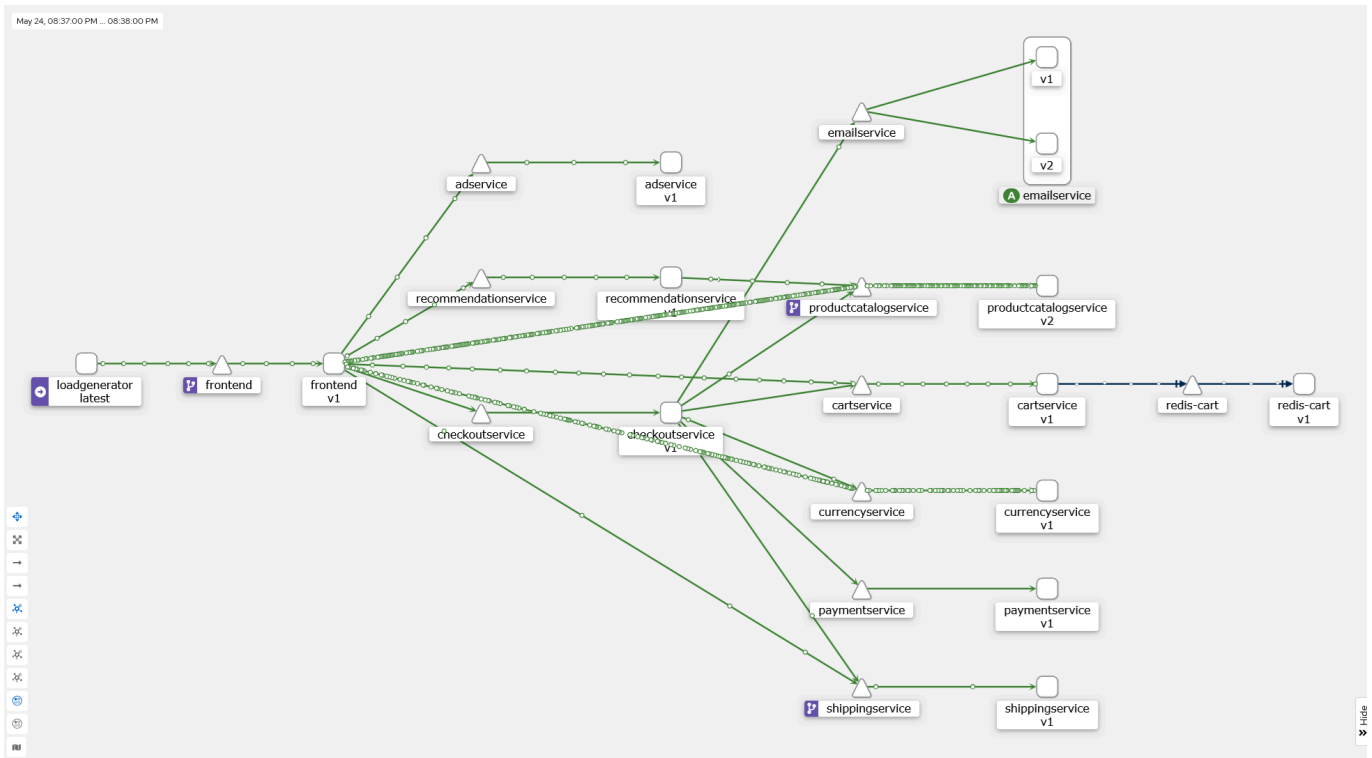
Task 6 – Traffic management, traffic routing

1. Apply DestinationRule and VirtualService policies to all services deployed in the service mesh regardless of how many versions each service provides. Route the traffic to version v1 of each service.

```
> kubectl -n default apply -f templates/release/istio/shippingservice-dr.yaml
> kubectl -n default apply -f templates/release/istio/shippingservice-vs.yaml
```

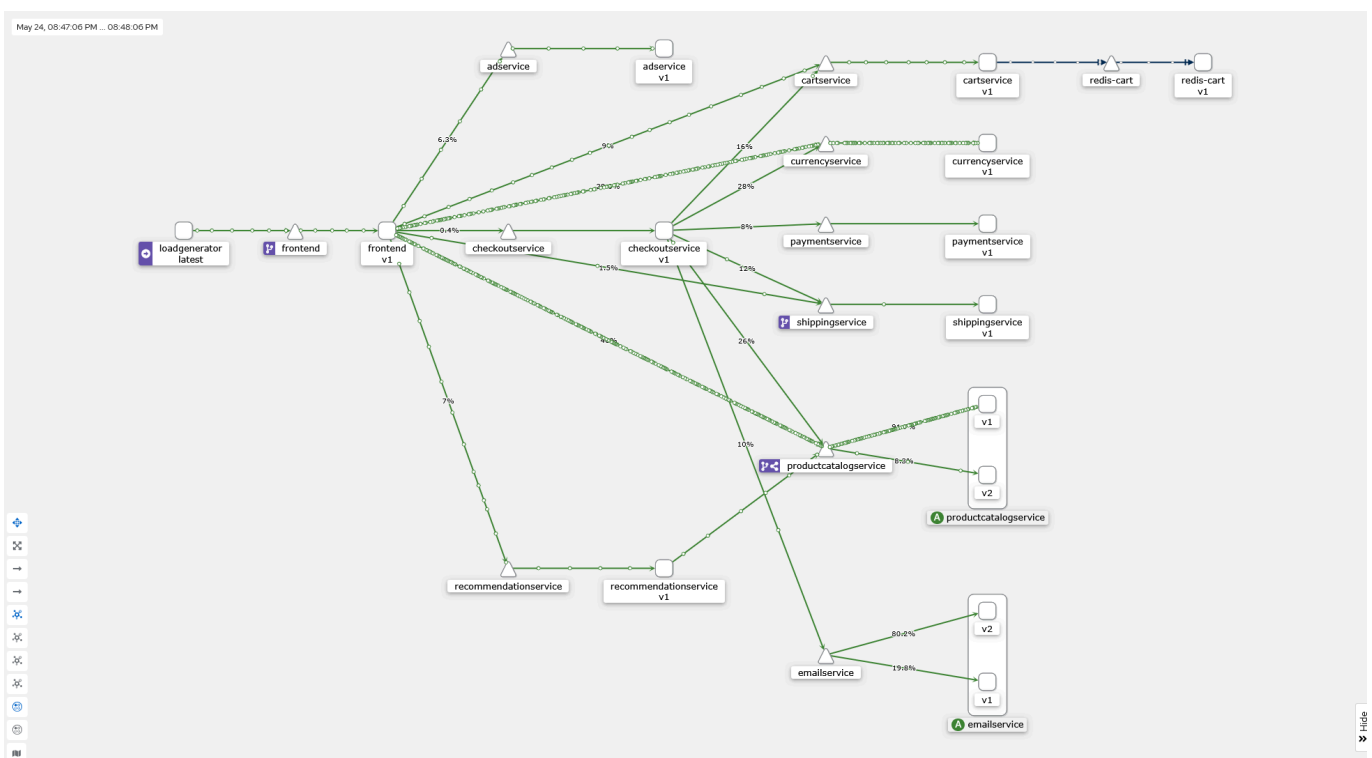
2. Route productcatalog traffic to version v2.

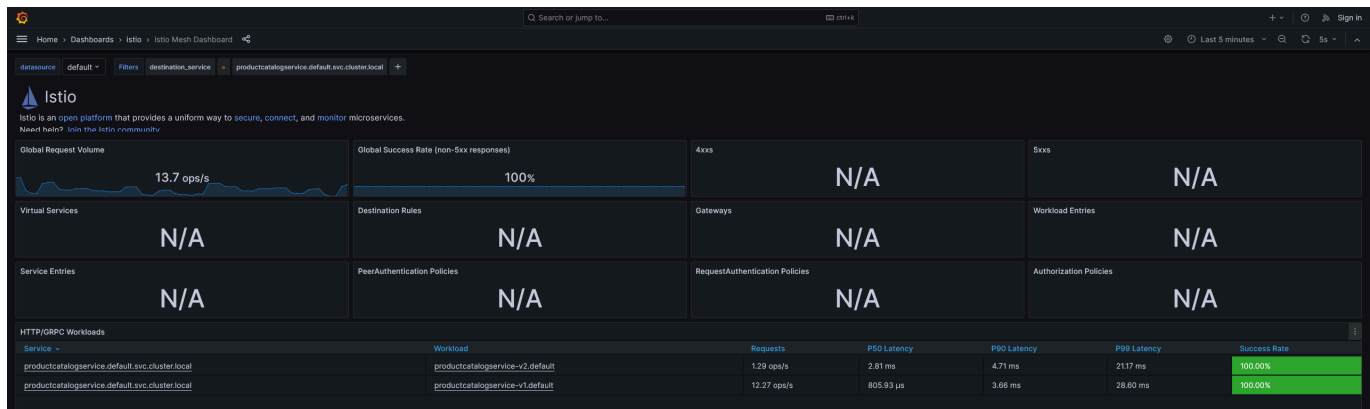
```
> kubectl -n default apply -f templates/release/istio/productcatalogservice-vs-3.yaml
```



Task 7 – Traffic management, traffic shifting

```
> kubectl -n default apply -f templates/release/istio/productcatalogservice-vs-1.yaml
```





```
> kubectl -n default apply -f templates/release/istio/productcatalogservice-vs-2.yaml
```

```
> kubectl -n default apply -f templates/release/istio/productcatalogservice-vs-3.yaml
```

Task 8 – Traffic management, traffic shifting



Uh, oh!

Something has failed. Below are some details for debugging.

HTTP Status: 500 Internal Server Error

```
rpc error: code = unknown desc = fault filter abort
could not retrieve products
main.(*frontendServer).homeHandler
/go/src/github.com/googlecloudplatform/microservices-demo/src/frontend/handlers.go:59
net/http.HandlerFunc.ServeHTTP
/usr/local/go/src/net/http/server.go:1995
github.com/googlecloudplatform/microservices-demo/src/frontend/vendor/github.com/gorilla/mux.(*Router).ServeHTTP
/go/src/github.com/googlecloudplatform/microservices-demo/src/frontend/vendor/github.com/gorilla/mux.go:212
main.(*logHandler).ServeHTTP
/go/src/github.com/googlecloudplatform/microservices-demo/src/frontend/middleware.go:181
main.ensureSessionInFunc1
/go/src/github.com/googlecloudplatform/microservices-demo/src/frontend/middleware.go:181
net/http.HandlerFunc.ServeHTTP
/usr/local/go/src/net/http/server.go:1995
github.com/googlecloudplatform/microservices-demo/src/frontend/vendor/go.opencensus.io/plugin/ochttp.(*Handler).ServeHTTP
/go/src/github.com/googlecloudplatform/microservices-demo/src/frontend/vendor/go.opencensus.io/plugin/ochttp/server.go:82
net/http.serverHandler.ServeHTTP
/usr/local/go/src/net/http/server.go:2774
net/http.(*conn).serve
/usr/local/go/src/net/http/server.go:1878
runtime.goroutine
/usr/local/go/src/runtime/asm_amd64.s:1337
```

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